MARYMOUNT
UNIVERSITY

RECEIPT SCANNER

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RECEIPT SCANNER

This document provides an overview of the Receipt Scanner application software requirements, design, and usage. This document is produced for IT-548-A, Marymount University, as a group project. Volume 4 of this document is produced and stored externally.

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Software Requirements Completion Date:	3/16/2026
Software Design Completion Date:	3/23/2026
Software Implementation Date:	3/23/2026
Software User Guide Completion Date	

ORGANIZATION

This document is organized into the following sections:

Volume-1: Receipt Scanner Planning Document

Volume-2: Receipt Scanner Software Requirements

Volume-3: Receipt Scanner Software Design Description

Volume-4: Receipt Scanner Implementation

Volume-5: Receipt Scanner Usage Guide

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SOFTWARE PLANNING

RECEIPT SCANNR VOLUME 1

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CHAPTER 1
PROJECT OVERVIEW

1.1 Purpose

The purpose of this project is to explore the use of modern software development tools and artificial intelligence techniques within a Python-based application to solve a practical real-world problem. Specifically, the system aims to demonstrate how Optical Character Recognition (OCR) and automated information extraction techniques can be used to digitize paper receipts and derive meaningful financial information from them.

Many individuals and small businesses rely on paper receipts to track expenses. Manual entry of receipt data into spreadsheets or financial systems is time-consuming and prone to human error. As businesses accumulate large numbers of receipts, organizing and analyzing expense information becomes inefficient.

The Receipt Scanner application will allow a user to upload images of receipts and automatically extract key fields such as transaction date, vendor name, price paid, and gallons purchased. The extracted data will then be organized into collections of receipts defined by the user.

Using these extracted values, the system will compute summary statistics for each collection, including the average price per gallon, the total amount spent, the number of receipts in the collection, and the date range represented by the receipts.

This project is intended as a proof-of-concept system designed to demonstrate the feasibility of combining OCR technology with simple automated parsing techniques in a Python environment. The application is not intended to serve as a production-ready financial system, but rather as a prototype illustrating how automation tools can assist small single-user businesses with basic expense analysis.

1.2 Roles

The development team consists of the following members:

James Green

Responsible for system planning, requirements development, implementation of core application functionality, and documentation of system behavior.

Chris Duckers

Responsible for system design development, review of requirements and implementation artifacts, and collaboration on application development.

Numi

Responsible for creating and maintaining documentation, presentation, and users guide.

All team members will participate in system testing, review activities, and documentation refinement throughout the project lifecycle.

1.3 Deliverables

The project will produce the following deliverables:

1.3.1 Planning Document

Provides an overview of the project goals, objectives, and development strategy. This document outlines the development timeline, team responsibilities, tools and technologies used, and project assumptions. It serves as the foundation for all subsequent development activities and ensures that the team has a clear understanding of the project scope and direction

1.3.2 Software Requirements Specification (SRS)

Defines the functional and non-functional requirements of the Receipt Scanner application. This document describes the expected behavior of the system, the features it must provide, and the constraints under which it must operate. The SRS serves as a formal agreement between developers and reviewers regarding what the system must accomplish.

1.3.3 Software Design Description (SWDD)

Outlines the architectural and technical design of the Receipt Scanner system. This document explains how the system components interact, the structure of the software modules, and the technologies used to implement the application. The design document provides detailed technical information necessary for implementing and maintaining the system.

1.3.4 Application Implementation

Consists of the fully developed Receipt Scanner software written in Python. The application will include modules responsible for image upload, OCR processing, data extraction, receipt organization, and statistical analysis.

1.3.5 User Guide

Provides instructions for installing, running, and using the Receipt Scanner system. It explains how users can upload receipt images, create receipt collections, view extracted information, and analyze summary statistics.

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CHAPTER 2
DEVELOPMENT

2.1 Assumptions

The following assumptions apply to the development and operation of the Receipt Scanner system:

- Users will provide receipt images in a clear and readable format suitable for OCR processing.
- Receipts are assumed to be standard retail receipts containing recognizable textual patterns such as vendor names, dates, and totals.
- The system will be used by a single user operating locally on their machine.
- Browser storage may impose limitations on the number of receipts that can be loaded simultaneously.
- Uploaded files will be validated to confirm they are image files before processing.
- Security concerns related to storing uploaded images locally are considered out of scope for this proof-of-concept implementation.
- Receipt collections will typically contain fewer than 100 receipts.

2.2 Tools and Languages

1. The Receipt Scanner system will be implemented entirely in Python.
2. The following technologies and tools will be used during development:
3. Flask python web framework will be used to host a local web server and provide the browser-based interface.
4. **OCR Processing**

   An open-source OCR engine such as **Tesseract OCR** will be used to extract text from uploaded receipt images.

5. **Image Processing**

   Python libraries such as **Pillow** or **OpenCV** may be used for image preprocessing prior to OCR extraction.

6. **Frontend Interface**

   The application will provide a simple browser-based interface allowing users to upload receipts, manage receipt collections, and view extracted data summaries.

7. Receipt images will be stored on the local file system, while extracted receipt metadata will be stored within the browser session for display and analysis.
8. All tools used in this project will be open source.

2.1 Version Control

Source code and documentation for the Receipt Scanner system will be maintained using a version control system such as Git. Version control will be used to track changes to the codebase, manage revisions to project documentation, and support collaborative development between team members.

CHAPTER 3
DEPLOYMENT

3.1 Hosting

Deployment of the ReceiptScanner webpapp will be hosted on Render.com through through the domain https://preview.ckplace.org

3.2 Environment

To facilitate deployment, a docker container containing the necessary installations will be utilized. The contents of this docker file are below:

```
FROM python:3.12-slim

ENV PYTHONDONTWRITEBYTECODE=1
ENV PYTHONUNBUFFERED=1
ENV PORT=10000

WORKDIR /app

RUN apt-get update && apt-get install -y --no-install-recommends \
    tesseract-ocr \
    tesseract-ocr-eng \
    && rm -rf /var/lib/apt/lists/*

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY . .
RUN pip install --no-cache-dir -e .

CMD ["python", "-m", "marymount.edu.receiptscanner.web"]
```

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SOFTWARE REQUIREMENTS

RECEIPT SCANNER VOL 2

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CHAPTER 4
INTRODUCTION

4.1 Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements of the Receipt Scanner application.

The purpose of this document is to define the behavior and capabilities of the system in a clear and structured manner so that developers and reviewers can understand the expected functionality of the application prior to implementation.

4.2 Scope

The Receipt Scanner system is a Python-based application designed to digitize receipt images and extract structured information from them using Optical Character Recognition (OCR).

The system allows a user to create named collections of receipts, upload receipt images, extract relevant transaction information, and compute summary statistics based on the extracted values.

The application will operate locally and will provide a browser-based interface for interacting with the system. The system is intended as a proof-of-concept prototype and is not designed for enterprise or multi-user deployment.

4.3 System Overview

The system provides a lightweight workflow for organizing and analyzing receipt data.

Users can create collections that represent groups of receipts. Receipts may then be uploaded individually or in batches. Each uploaded image will be processed using OCR to extract textual content. The system will then attempt to identify specific receipt fields such as vendor name, transaction date, price paid, and gallons purchased.

After extraction, the system will display the extracted information and allow the user to manually edit any fields if the automatic extraction is incorrect. Receipts flagged as problematic by the extraction process will be displayed separately and excluded from summary calculations.

The system will compute several summary statistics for each collection, including:

- Average price per gallon
- Total amount spent
- Date range of receipts
- Number of valid receipts in the collection

These summaries allow users such as small single-operator businesses to quickly analyze vehicle fuel expenses.

CHAPTER 5
FUNCTIONAL REQUIREMENTS

5.1 Upload Receipt Image

Table R4-1 Functional Requirements

FR1-1	Receipt Scanner **shall** support the following image formats ➢ JPG ➢ PNG ➢ HEIC
FR1-2	Receipt Scanner **shall** process each receipt individually.

5.2 Store Uploaded File

Table R4-2 Functional Requirements

FR2-1	Receipt Scanner **shall** store uploaded receipt images.
FR2-2	Receipt Scanner **shall** verify that uploaded files are valid image formats.

5.3 Perform OCR

Table R4-3 Functional Requirements

FR3-1	Receipt Scanner **shall** perform Optical Character Recognition on each uploaded receipt image.

5.4 Extract Receipt Fields

Table R4- Functional Requirements

FR4-1	Receipt Scanner **shall** attempt to identify the fields from the OCR output: ➢ Vendor name ➢ Transaction date ➢ Total price paid ➢ Gallons purchased

5.5 Display Extracted Results

Table R5- Functional Requirements

FR5-1	Receipt Scanner **shall** display the extracted receipt information to the user.
FR5-2	Receipt Scanner **shall** display summary statistics for the receipt collection.

5.6 Allow Manual Correction

Table R4-6 Functional Requirements

FR6-1	Receipt Scanner **shall** allow the user to manually edit extracted receipt fields.

5.7 Save or Export Results

Table R4-7 Functional Requirements

FR7-1	Receipt Scanner **shall** retain receipt data within the application interface for the current session.

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CHAPTER 6
INTERFACE REQUIREMENTS

6.1 User Interface

Table R5-1 Interface Requirements

IR1-1	Receipt Scanner **shall** provide a browser-based interface allowing users to: ➢ Upload receipt images ➢ View extracted receipt data ➢ Edit extracted fields ➢ View summary statistics

6.2 Software Interfaces

Table R5-2 Interface Requirements

IR2-1	Receipt Scanner **shall** interact with an OCR engine to convert receipt images into machine-readable text.

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CHAPTER 7
NON-FUNCTIONAL REQUIREMENTS

7.1 Usability

Table R6-1 Non-Functional Requirements

NFR-1-1	Receipt Scanner **shall** provide a simple and intuitive web interface suitable for users with minimal technical experience.

7.2 Reliability

Table R6-2 Non-Functional Requirements

NFR-2-1	Receipt Scanner **shall** identify receipts that fail field extraction and flag them as problematic rather than using them in summary calculations.

7.3 Maintainability

Table R6-3 Non-Functional Requirements

NFR-3-1	Receipt Scanner **shall** be implemented in modular Python components to allow future improvements to OCR processing or field extraction logic.

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CHAPTER 8
ACADEMIC REQUIREMENTS

8.1 Assignment Requirements

Table R7-1 Academic Requirements

AR-5-1	Receipt Scanner **shall** utilize Natural Language Processing, AI, or Intelligent Image Processing.
AR-5-2	Each Group Member **shall** participate in the coding process.

8.2 External Requirements

Table R7-1 Academic Requirements

AER-5-1	All external libraries utilized will be specifically called out and included on the presentation
AER-5-2	Documentation for the project **shall** include all aspects of planning, documentation, and usability.

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SOFTWARE DESIGN

RECEIPT SCANNER VOL 3

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CHAPTER 1
INTRODUCTION

1.1 Purpose

This Software Design Description (SWDD) defines the internal architecture and design of the Receipt Scanner system. The purpose of this document is to describe how the system is organized, how receipt data is processed, and how the main components interact to support upload, OCR extraction, field parsing, review, correction, and summary display.

1.2 Scope

This document describes the design of the Receipt Scanner software system, which is implemented using Python. The design focuses primarily on the internal processing engine responsible for receipt analysis and data extraction.

The system supports the following primary capabilities:

- ingestion of receipt images
- OCR extraction of text from images
- parsing of receipt fields
- validation and flagging of problematic receipts
- organization of receipts into collections
- generation of summary statistics for collections

External interfaces such as browser-based frontends are treated as adapters that invoke the core Python services. The SWDD therefore focuses on the design of the internal processing components rather than the implementation of any particular user interface.

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CHAPTER 2
SYSTEM ARCHITECTURE

The receipt Scanner system design uses a modular architecture that allows the separation of the receipt's core functionality from the user interface. This allows the system to solely focus on the core problem which is extracting structured data from the receipt images while also being flexible in how the results are used.

The system architecture consists of the following components who will perform a specific task in the receipt processing pipeline:

- Image input Module
- OCR Processing Module
- Text Processing and Classification Module
- Data Structuring Module
- Interface Layer

The modular design ensure that the main processing engine will function independently from the interface used to display the results. For example, the structured data receipt can be presented via web interface, exported to a file for further analyst or presented via command line interface.

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CHAPTER 3
SYSTEM DESIGN OVERVIEW

3.1 Overview

The Receipt Scanner is a modular Python application that extracts structured information from receipt images. The current implementation supports both a Flask-based web interface and a command-line interface, both of which use the same receipt-processing pipeline.

At a high level, the system receives a receipt image, preprocesses it, extracts text through OCR, parses key fields from the text, stores the result in memory, and presents the result for review and correction.

3.2 Architectural Style

The application follows a lightweight layered design combined with a sequential processing pipeline. These layers include:

- **Presentation Layer** for Flask routes and CLI argument handling
- **Service Layer** for receipt records, status values, and summaries
- **Processing Layer** for image preprocessing, OCR extraction, and field parsing
- **Infrastructure Layer** for local file storage and external dependencies

3.3 Major Subsystems

3.3.1 Web Interface

This web interface supports receipt upload, receipt listing, manual correction, processed-image preview, original-file download, and summary display. Collections are also provided by this interface. This interface interacts with the service layer using the ReceiptScanner API.

3.3.2 Command Line Interface

A basic CLI that uses receipt scanner API and JSON output to extract data from receipts.

3.3.3 Processing Pipeline

This pipeline is provided by the core ReceiptScanner publicly available object. The receipt scanner. This pipline contains the image preprocessor, the OCRProcessor, the Text processor, and the container object.

3.3.4 Service Layer

This subsystem acts as an in-memory receipt store for the web-interface.

CHAPTER 4
COMPONENT DESIGN

4.1 Purpose

This chapter describes the main components of the current Receipt Scanner implementation and their responsibilities.

4.2 Presentation Components

4.2.1 Wb Application

The Flask web application is the primary interactive interface. It accepts uploaded files, invokes the scanner pipeline, stores and retrieves receipt records through the service layer, renders receipt lists and item views, supports manual correction, and serves previews and downloads.

4.2.2 Command-Line Interface

The CLI provides a non-web entry point into the same processing pipeline. It accepts image paths, creates a configured ReceiptScanner, processes one or more files, and returns JSON results or error payloads.

4.3 Service Components

ReceiptService maintains the in-memory receipt store and serves as the main state-management component of the application. It creates receipt records, stores parsed results, updates status values, marks records as fixed after manual correction, clears working data, and produces summary values.

4.4 Processing Components

4.4.1 ReceiptScanner

ReceiptScanner coordinates the main processing pipeline. It invokes image preprocessing, OCR extraction, and text parsing in sequence, then returns the final result. It is the main integration point used by both the web application and the CLI.

4.4.2 ImagePreprocessor

ImagePreprocessor validates and prepares uploaded images for OCR. It checks file existence and supported type, handles HEIC restrictions, converts the image to grayscale, applies simple contrast normalization, and saves a processed image file.

Input: raw image file path **Output:** processed image file path

4.4.3 OCRProcessor

OCRProcessor performs text extraction from the processed image using pytesseract and Tesseract. It opens the processed image, runs OCR, cleans the extracted text, rejects empty results, and writes OCR logs for debugging.

Input: processed image file path **Output**: cleaned OCR text

4.4.4 TextProcessor

TextProcessor converts OCR text into structured receipt data. It identifies a merchant using a predefined vendor alias list, extracts the total and date, extracts gallons, calculates price per gallon, and preserves OCR lines as supporting output.

Input: OCR text **Output**: parsed receipt result dictionary

This component is rule-based and depends heavily on OCR quality and expected text patterns. In the current implementation, missing total or gallons values can cause failure during price-per-gallon calculation.

4.5 Supporting Transformation

The web interface also uses receipts_dataframe() to convert the raw store into a UI-ready structure. This helper carries forward key fields, normalizes selected values, and computes whether a record should be treated as broken by the interface.

4.6 Component Relationships

The main relationships are:

- The web application and CLI call ReceiptScanner
- ReceiptScanner uses ImagePreprocessor, OCRProcessor, and TextProcessor
- The web application stores and retrieves records through ReceiptService
- Receipts_dataframe() converts stored records into a structure suitable for display and summary calculations

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CHAPTER 5
DATA DESIGN

5.1 Purpose

This chapter describes the main data structures used by the current Receipt Scanner implementation.

5.2 Input Data

The application accepts receipt image files through the web interface and the CLI. Supported input types include JPG, JPEG, PNG, and conditionally HEIC when optional support is installed. The web interface also accepts manual correction input for fields such as merchant, date, total, gallons, price per gallon, and lines.

5.3 In-Memory Receipt Store

The main application store is an in-memory dictionary managed by ReceiptService. Each key is a receipt identifier, and each value is a receipt record.

A receipt record generally includes fields such as:

```
{
    "<uid>": {
        "filename": str,
        "status": str,
        "result": dict | None,
        "path": str,
        "fixed": bool,
        "collection": str | None,
    }
}
```

This represents the normal store shape rather than a strict schema for every record. Some sample or error records may omit fields shown in the normal runtime structure.

5.4 Parsed Result Data

A successful parsed result generally includes:

- merchant
- date
- total
- gallons
- gallons_source
- price_per_gallon
- price_per_gallon_source
- lines

When processing fails, the result may instead contain an error field. As with the receipt store itself, this is a general result shape rather than a guaranteed fixed schema.

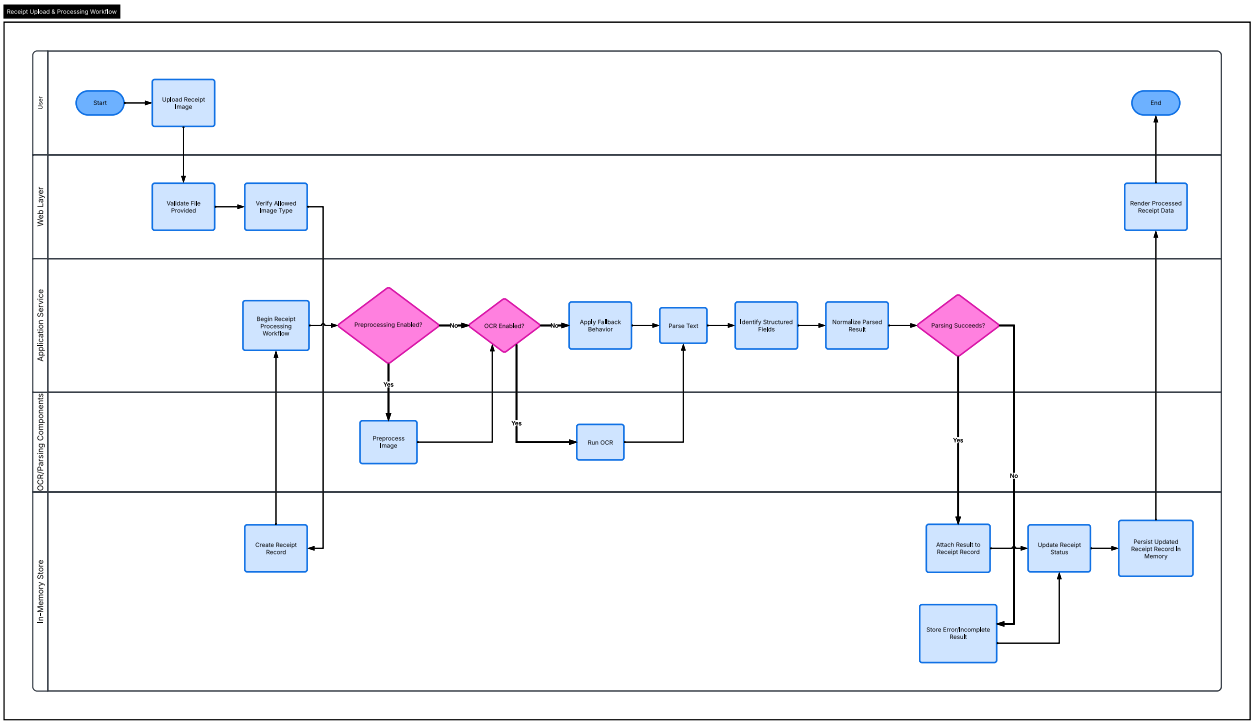
5.5 Data Quality and Derived Data

The service layer partially normalizes stored results. In particular, gallons and price_per_gallon are normalized more consistently than total, which may remain a string until later display or summary processing. The current design therefore uses mixed data typing in some areas.

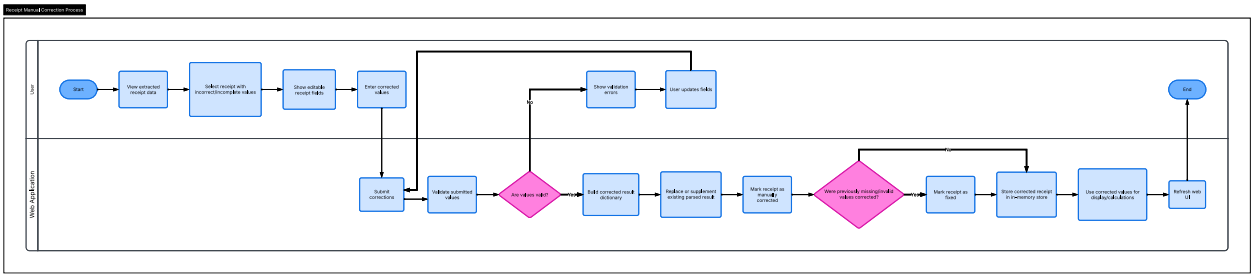
The web application also uses receipts_dataframe() to derive a UI-ready structure from the raw store. This derived view carries forward key fields and computes a broken flag used by the interface. Sample receipts are preloaded at startup and restored when the working dataset is cleared.

CHAPTER 6
PROCESSING WORKFLOWS

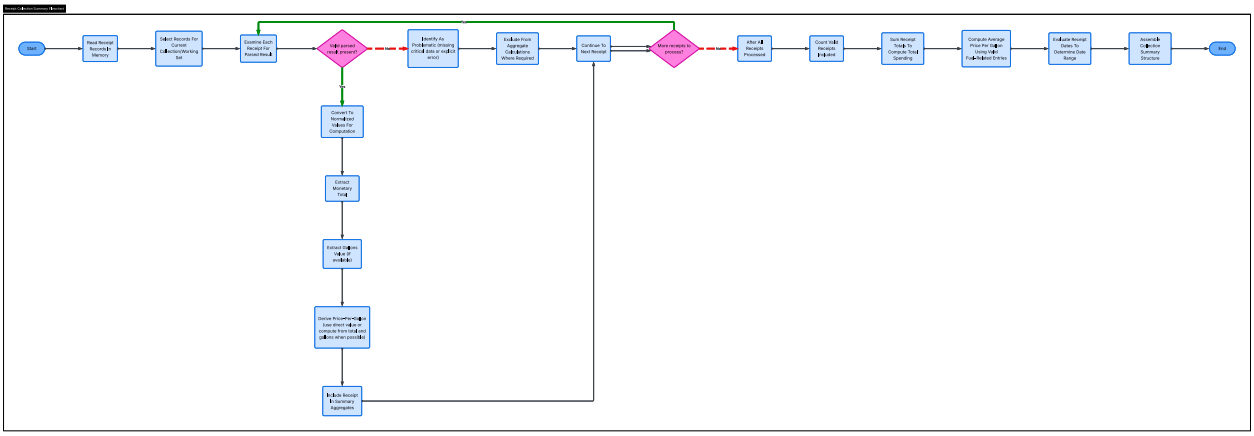
6.1 Single Receipt Processing Flow



6.2 Manual Correction Flow



6.3 Summary Generation Flow



CHAPTER 7
ERROR HANDLING AND CONSTRAINTS

7.1 Purpose

This chapter describes how the current Receipt Scanner implementation handles failures and what practical constraints affect its operation.

7.2 Error Handling Approach

The application uses a simple exception-based error handling model. Failures are generally detected inside preprocessing, OCR, parsing, or file-handling operations and then surfaced to the caller.

In the web interface, processing failures are stored in the receipt record and shown in the application. In the CLI, failures are returned as error payloads in JSON output. This approach is appropriate for the current prototype because it keeps failures visible without introducing complex recovery logic.

7.3 Common Failure Cases

Common failure cases include:

- missing upload file or missing CLI input
- unsupported image type
- missing file on disk
- image-open failure in Pillow
- HEIC input without HEIC support installed
- OCR dependency failure
- empty OCR output
- parsing failure when expected values are missing
- missing processed preview file
- missing original file for download

The most important current parsing failure occurs when TextProcessor attempts to calculate price per gallon even though total or gallons was not extracted successfully.

7.4 Key Constraints and Known Weaknesses

The current design has several important constraints:

- OCR accuracy depends heavily on image quality
- the parser is rule-based and sensitive to receipt format
- merchant detection depends on a predefined vendor alias list
- processing runs synchronously during web requests
- receipt data is stored only in memory
- stored values use mixed data typing in some areas

There is also a small inconsistency in how record validity is judged. The interface uses broken-state logic derived from receipts_dataframe(), while service-level summaries rely more directly on whether totals are present and usable.

CHAPTER 8
DESIGN VS IMPLENTATION CONSIDERATIONS

The current implementation of the Receipt Scanner system reflects a proof-of-concept realization of the proposed architecture. While the design specifies a clear separation between interface, service, and infrastructure layers, the implementation consolidates some of these responsibilities to simplify development and deployment.

Key deviations include:

- Partial integration of service-layer logic within the web application
- Use of lightweight in-memory data structures instead of formal data models
- Simplified module boundaries to prioritize rapid prototyping

Despite these simplifications, the system preserves the core architectural intent and demonstrates the feasibility of a modular receipt processing pipeline. The design supports future refactoring to fully align with the intended layered architecture.

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USER GUIDE

RECEIPT SCANNER VOL 5

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CHAPTER 1
INTRODUCTION

1.1 Purpose

The purpose of this User Guide is to provide clear and detailed instructions for installing, configuring, and using the Receipt Scanner application. This guide is intended to help users understand how to interact with the system, upload receipt images, view extracted data, and analyze receipt collections.

The User Guide is designed for users with minimal technical experience and provides step-by-step instructions to ensure that the application can be used effectively without requiring advanced knowledge of programming or software systems.

1.2 Intended Users

The Receipt Scanner application is designed for a wide range of users who need a simple and efficient way to digitize and analyze receipt data. Intended users include:

- Students working on academic projects involving data extraction and analysis
- Small business owners tracking expenses such as fuel purchases
- Individuals managing personal finances and receipts
- Users with limited technical background who require an easy-to-use interface

1.3 Overview of the System

The Receipt Scanner system allows users to upload images of receipts and automatically extract key information using Optical Character Recognition (OCR). The system processes the receipt images and identifies important fields such as vendor name, transaction date, total price, and gallons purchased.

The extracted data is then organized into a structured format and displayed to the user. Users can review and edit the extracted information if necessary and view summary statistics for their receipt collections.

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CHAPTER 2
SYSTEM REQUIREMENTS

To properly run an instance of Receipt Scanner Locally, the following requirements must be met:

Tesseract OCR engine installed

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CHAPTER 3
INSTALLATION INSTRUCTIONS

To run Receipt Scanner locally, perform the following steps to install the software:

Step 1: Install Python

Download and install Python 3.x from the official Python website if it is not already installed on your system.

Step 2: Install Required Libraries

Open a terminal or command prompt and run the following command:

pip install -r requirements.txt

Step 3: Install Tesseract OCR

Download and install the Tesseract OCR engine for your operating system.

After installation, ensure that the Tesseract executable is added to your system's PATH environment variable so that it can be accessed by the application.

Step 4: Download Project Files

Download or clone the Receipt Scanner project files to your local machine.

Step 5: Verify Installation

Ensure that all dependencies are installed correctly by running the application and checking for errors.

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CHAPTER 4
PROGRAM EXECUTION

4.1 Starting the Application

If accessing the application via the web, no installation is required to start. Simply navigate to https://preview.ckplace.org.

If accessing the application via local use, after performing the installation, launch the application via python -m Marymount.edu.receiptscanner.web.py.

4.2 Using the System

4.2.1 Uploading a Receipt Image

To upload a receipt image:

1. Click on the “Upload Receipt” button
2. Select an image file from your computer
3. Click “Submit”

The system will automatically process the receipt and extract text using OCR.

4.2.2 Viewing Receipt Data

After processing, the system will display the extracted receipt information. This typically includes:

- Vendor name
- Transaction date
- Total price
- Gallons purchased

The extracted data is presented in a structured format for easy reading.

4.2.3 Manual Edits

Users can manually edit the values.

To edit data:

1. Click on the field you wish to modify
2. Enter the correct value
3. Save the changes

This ensures that inaccurate OCR results can be corrected before analysis.

4.2.4 Managing Receipt Collections

Users can organize receipts into collections for easier management.

- Collections allow users to:
- Group related receipts
- View multiple receipts together
- Perform analysis on grouped data

4.2.5 Viewing Summary Statistics

The system automatically calculates summary statistics for receipt collections, including:

- Total amount spent
- Average price per gallon
- Number of receipts
- Date range of transactions

These statistics help users quickly analyze spending patterns.

4.3 Error Messages

The following error messages can appear in the program.

Table U4-3 Error Messages

ERROR MESSAGE	CAUSE	SOLUTION
OCR not working	Tesseract not installed	Install Tesseract OCR and verify PATH
Incorrect data extraction	Poor image quality	Upload a clearer image or edit manually
Application not starting	Missing dependencies	Reinstall required Python libraries
Image upload fails	Unsupported format	Use JPG, PNG, or HEIC format

4.4 Best Practices

To achieve the best results when using the Receipt Scanner system, users should follow these guidelines:

- Use clear, high-resolution images of receipts
- Ensure receipts are well-lit and not blurry
- Avoid folded or damaged receipts
- Review extracted data for accuracy

Following these practices will improve OCR accuracy and overall system performance.

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APPENDIX A

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NOMENCLATURE

Application Service Layer	The logical processing layer that coordinates OCR, parsing, and data handling.
Flask	A lightweight Python web framework used to implement the application interface.
Gallons Purchased	The quantity of fuel purchased, used for computing price per gallon.
Interface Layer	The component responsible for user interaction (web UI).
Infrastructure Layer	Supporting services such as OCR engines and file handling systems.
Manual Correction	The process by which users adjust incorrectly extracted data fields.
OCR (Optical Character Recognition)	A technology used to convert images of text into machine-readable text.
Parsed Data	Structured data extracted from OCR output, including fields such as vendor, date, and total.
Parsing	The process of analyzing OCR text to extract structured fields using rules or patterns.
Receipt Collection	A user-defined grouping of receipts used for aggregation and analysis.
Receipt Scanner	The software system developed in this project to process receipt images and extract structured data.

Regex (Regular Expression)	A pattern-matching technique used to identify structured data within text.
Session Storage	Temporary storage of data during runtime, not persisted after application termination.
Summary Statistics	Aggregated values calculated from a collection of receipts (e.g., total spend, average price).
Tesseract OCR	An open-source OCR engine used to extract text from receipt images.
Vendor	The merchant or business from which a purchase was made.

APPENDIX B

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EXTERNAL LIBRARIES

argparse	Parses command-line arguments and enables optional CLI-based interaction with the application (e.g., running in different modes or configurations).
Flask	Provides the browser-based interface and handles HTTP requests, routing, and rendering.
json	Provides functionality for encoding and decoding data in JSON format for structured data handling and potential export/import operations.
pathlib	Provides cross-platform file path handling and file management.
Pillow	Used for image manipulation and preprocessing prior to OCR.
re	Text Parsing/Pattern Matching
sys	Provides access to system-level parameters and runtime environment variables, including command-line arguments and execution control.
Tesseract OCR	Extracts text from receipt images. Required for full functionality.
typing	Supports static type annotations in Python to improve code readability, maintainability, and developer tooling support.

uuid	Identifier Generation

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